

# Complex Number

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# 1 Complex Number

## I Complex number (複數)

### i Complex Number

The set of complex numbers  $\mathbb{C}$  is defined as

$$\mathbb{C} := \{a + bi \mid a, b \in \mathbb{R}\},$$

where  $i$  is defined as the only complex number such that  $i^2 = -1$  and is called the imaginary unit.

For a complex number  $z = a + bi$  with  $a, b \in \mathbb{R}$ ,  $a$  is called its real part (實部) and denoted as  $\Re(z)$ ,  $\text{Re}(z)$ , or  $\mathcal{R}e(z)$ ,  $b$  is called its imaginary part (虛部) and denoted as  $\Im(z)$ ,  $\text{Im}(z)$ , or  $\mathcal{I}m(z)$ ,  $z$  is called an imaginary number (虛數) if  $b \neq 0$ , and  $z$  is called a pure imaginary number (純虛數) if  $a = 0$  and  $b \neq 0$ .

### ii Addition, subtraction, and multiplication

For two complex numbers  $y = a + bi$  and  $z = c + di$  with  $a, b, c, d \in \mathbb{R}$ :

$$y + z = (a + c) + (b + d)i.$$

$$y - z = (a - c) + (b - d)i.$$

$$y \cdot z = (ac - bd) + (ad + bc)i.$$

### iii Absolute value (絕對值), magnitude, radial distance (向徑), or modulus (模/模長)

The absolute value, magnitude, radial distance, or modulus of a complex number  $z = a + bi$  with  $a, b \in \mathbb{R}$  is  $|z| = \sqrt{a^2 + b^2}$ .

### iv Complex conjugate (共軛)

The (complex) conjugate of a complex number  $z = a + bi$  with  $a, b \in \mathbb{R}$ , denoted as  $\bar{z}$  or  $z^*$  and called  $z$  bar, is defined as  $a - bi$ .

### v Division

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}, z \neq 0$$

### vi Properties

For complex numbers  $z, z_1, z_2$ :

$$|z| = \sqrt{z \cdot \bar{z}}$$

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2.$$

$$\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2.$$

$$\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}.$$

$$\Re(\overline{z_1} \cdot z_2) = \Re(z_1 \cdot \overline{z_2}).$$

$$\frac{\overline{z_1}}{\overline{z_2}} = \overline{\left(\frac{z_1}{z_2}\right)}, \quad z_2 \neq 0.$$

$$\overline{z^n} = (\overline{z})^n, \quad z \neq 0 \wedge n \in \mathbb{Z}.$$

## II Polynomials

### i Fundamental theorem of algebra (代數基本定理), d'Alembert's theorem, or the d'Alembert–Gauss theorem

Every non-constant single-variable polynomial with complex coefficients has at least one complex root.

Every non-zero, single-variable, degree  $n$  polynomial with complex coefficients has, counted with multiplicity, exactly  $n$  complex roots.

### ii Equality of polynomials with real coefficients of complex number and its conjugate

For all polynomials with real coefficients  $f$ , for all complex numbers  $z$ :

$$\overline{f(z)} = f(\overline{z}).$$

Pair of imaginary roots (虛根成對定理): If a polynomial with real coefficients has an imaginary root, then its conjugate is also a root. Thus, an odd-degree polynomial with real coefficients must have at least one real root.

## III Argument

### i Argument (輻角)

For a complex number  $z \neq 0$ , the argument of  $z$ , denoted as  $\arg(z)$ , is defined as any real number  $\varphi$  such that

$$z = |z|e^{i\varphi}.$$

### ii Principal argument (輻角主值/主輻角)

For a complex number  $z \neq 0$ , the principal argument of  $z$ , denoted as  $\text{Arg}(z)$ , is defined as the argument of  $z$  with in  $(-\pi, \pi]$ . (Some sources define it as the argument of  $z$  with in  $[0, 2\pi)$ .)

### iii Polar form (極式)

The polar form of a complex number  $z$  is

$$z = |z|(\cos \theta + i \sin \theta), \quad \theta \in \mathbb{R}.$$

#### iv Properties

For any real number  $\theta, \varphi, a, b$ :

$$\overline{e^{i\theta}} = e^{-i\theta}.$$

$$(ae^{i\theta})(be^{i\varphi}) = (ab)e^{i(\theta+\varphi)}.$$

$$\frac{ae^{i\theta}}{be^{i\varphi}} = \frac{a}{b}e^{i(\theta-\varphi)}.$$

### IV Complex plane (複數平面), Argand plane (阿爾岡平面), or Gaussian plane (高斯平面)

#### i Complex plane, Argand plane, or Gaussian plane

The complex plane is the plane formed by the complex numbers, with a Cartesian coordinate system such that the horizontal x-axis, called the real axis, is formed by the real numbers, and the vertical y-axis, called the imaginary axis, is formed by the imaginary numbers.

#### ii De Moivre's formula (隸美弗公式)

$$(r(\cos \theta + i \sin \theta))^n = r^n(\cos(n\theta) + i \sin(n\theta)), \quad r \neq 0 \wedge n \in \mathbb{Z}.$$

#### iii Hyperbolic De Moivre's formula

$$(r(\cosh \theta + i \sinh \theta))^n = r^n(\cosh(n\theta) + i \sinh(n\theta)), \quad r \neq 0 \wedge n \in \mathbb{Z}.$$

#### iv Dot product

The dot product of two vectors in  $\mathbb{R}^2$  corresponding to the two complex numbers  $z_1$  and  $z_2$  on the complex plane is  $\Re(z_1 \cdot \overline{z_2})$ .

### V Geometric meaning of exponentiations of complex numbers on complex plane

#### i Raciporial of positive number exponentiation of 1

For  $n \in \mathbb{N}$ ,  $1^{\frac{1}{n}}$ , that are, the  $n$  roots of  $x^n = 1$ , that are,  $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$  to the power of 0 to  $(n-1)$ , that are,

$$1^{\frac{1}{n}} = e^{i\frac{2\pi k}{n}}, \quad k \in \mathbb{N}_0 \wedge k < n,$$

are the  $n$  vertices of a regular  $n$ -gon inscribed in a unit circle centered at the origin including 1 on the complex plane.

#### ii Raciporial of positive number exponentiation of nonzero complex number

For  $n \in \mathbb{N}$  and  $z \in \mathbb{C}_{\neq 0}$ ,  $z^{\frac{1}{n}}$ , that are, the  $n$  roots of  $x^n = z$ , that are,  $\sqrt[n]{|z|}$  multiplied by  $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$  to the power of 0 to  $(n-1)$ , that are,

$$z^{\frac{1}{n}} = \sqrt[n]{|z|}e^{i\frac{\text{Arg}(z)+2\pi k}{n}}, \quad k \in \mathbb{N}_0 \wedge k < n,$$

are the  $n$  vertices of a regular  $n$ -gon inscribed in a circle of radius  $\sqrt[n]{|z|}$  centered at the origin including  $\sqrt[n]{|z|}e^{i\frac{\text{Arg}(z)}{n}}$  on the complex plane.

### iii Nonzero complex number exponentiation of nonzero complex number

For  $w, z \in \mathbb{C}_{\neq 0}$ ,  $z^w$ , that are,

$$\begin{aligned} z^w &= e^{w \ln(z)} \\ &= \left( |z| e^{i(\text{Arg}(z) + 2\pi k)} \right)^w \\ &= |z|^{\Re(w)} |z|^{\Im(w)i} e^{i\Re(w)(\text{Arg}(z) + 2\pi k)} e^{-\Im(w)(\text{Arg}(z) + 2\pi k)} \\ &= |z|^{\Re(w)} e^{-\Im(w)(\text{Arg}(z) + 2\pi k) + i(\Re(w)(\text{Arg}(z) + 2\pi k) + \Im(w) \ln|z|)}, \\ &k \in \mathbb{Z}, \end{aligned}$$

are points with equal angle difference  $\Re(w)2\pi$  on the spiral

$$|z|^{\Re(w)} e^{-\Im(w)\theta} \left( \cos(\Re(w)\theta + \Im(w) \ln|z|) + i \sin(\Re(w)\theta + \Im(w) \ln|z|) \right)$$

of parameter  $\theta \in \mathbb{R}$  including that of  $\theta = \text{Arg}(z)$  on the complex plane.

#### Modulus:

$$|z^w| = |z|^{\Re(w)} e^{-\Im(w)(\text{Arg}(z) + 2\pi k)}.$$

- When  $\Im(w) > 0$ : The modulus decays exponentially with respect to  $k$ , approaching zero as  $k$  increases and diverging to infinity as  $k$  decreases.
- When  $\Im(w) < 0$ : The modulus increases exponentially with respect to  $k$ , diverging to infinity as  $k$  increases and approaching zero as  $k$  decreases.
- When  $\Im(w) = 0$ : The modulus is always  $|z|^{\Re(w)}$ , lying on a circle with radius  $|z|^{\Re(w)}$  centered at the origin.

#### Angle (parameter $\theta$ ):

$$\theta(k) = \Re(w)(\text{Arg}(z) + 2\pi k) + \Im(w) \ln|z|.$$

- When  $\Re(w) > 0$ : The angle increases linearly with respect to  $k$ , rotating counterclockwise as  $k$  increases and clockwise as  $k$  decreases.
- .When  $\Re(w) < 0$ : The angle decreases linearly with respect to  $k$ , rotating clockwise as  $k$  increases and counterclockwise as  $k$  decreases.
- When  $\Re(w) \in \mathbb{Z}$ : The angle is always  $(\Re(w) \text{Arg}(z) + \Im(w) \ln|z|) \bmod (2\pi)$ , lying on the ray with angle  $\Re(w) \text{Arg}(z) + \Im(w) \ln|z|$ .

#### Spiral direction:

- When  $\Re(w)\Im(w) > 0$ : The root lies in a clockwise spiral centered at the origin on the complex plane.
- When  $\Re(w)\Im(w) < 0$ : The root lies in a counterclockwise spiral centered at the origin on the complex plane.

- When  $\Im(w) = 0$ : The root lies in a circle centered at the origin on the complex plane.

#### Number of distinct roots:

- When  $\Im(w) = 0 \wedge \Re(w) \in \mathbb{Q}$ : Suppose  $|\Re(w)| = \frac{m}{n}$  with  $n \in \mathbb{N}$  and  $\gcd(m, n) = 1$ , there are  $n$  distinct roots, which are the  $n$  vertices of a regular  $n$ -gon inscribed in a circle of radius  $|z|^{\Re(w)}$  centered at the origin including  $|z|^{\Re(w)} e^{i\Re(w)\text{Arg}(z)}$  on the complex plane.
- Otherwise: There are countably infinitely many distinct roots.

## VI Extended complex number

$$\{\bar{\mathbb{C}} = \{a + bi \mid a, b \in \bar{\mathbb{R}}\}.$$